FIT5145-Assignment 2

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GENERATIVE AI DECLERATION:

In this assignment, I used Gemini Chatbot to brainstorm, improve writing and clarity, explain the coding, concepts, and suggest how to approach the implementation.

Import data set and load library

```
# Load tidyverse
library(tidyverse)
```

```
# import the data set
#NOTE: When download the file, please select the "Download all data".
#https://ourworldindata.org/grapher/number-of-internet-users?time=earliest..2021
internet_users <- read.csv('number-of-internet-users.csv')
# https://ourworldindata.org/grapher/annual-working-hours-per-worker?time=1998..latest&countr
y=~IRL#all-charts
working_hour <- read_csv('annual-working-hours-per-worker.csv')
ireland_news <- read_csv('ireland_news.csv')
```

Preparaton session:

```
# Convert the table to long form and tidy the data
#Tidy in the head_category columns for question 1
clean_ireland_news <- ireland_news %>%   separate_longer_delim(headline_category, delim=".")%
>%

    # These line of code will format the data at the head_category for question 2b.
  mutate(headline_category = str_to_lower(headline_category),
         headline_category = str_replace_all(headline_category, "_", "-"),
         headline_category = str_replace_all(headline_category, "&", "and"),
         headline_category = na_if(headline_category, ""),
        headline_category = na_if(headline_category, " ")) %>%

  # Ignore all N/A data for question 2a
  filter(
  !is.na(news_provider),
  !is.na(engagement_score),
  !is.na(headline_category),
  !is.na(headline_text),
  !is.na(publish_date),
  # It will ignore those that has "..."
  !str_detect(news_provider, fixed("...")),
  !str_detect(engagement_score, fixed("...")),
  !str_detect(headline_category, fixed("...")),
  !str_detect(headline_text, fixed("...")),
  !str_detect(publish_date, fixed("...")),
  str_length(headline_category) > 1,
   # Ignore those english letters is smaller than 1 and the wrong typo ones
  headline_category != "x86%",
  # Ignore those not in this range
  engagement_score >= 0 & engagement_score <= 1) %>%

  # Break date column into day of the week / date / month / year for question 4 and 7
  mutate(publish_date = dmy(publish_date),
    day_name = wday(publish_date, label = TRUE, abbr = FALSE),
    date     = day(publish_date),
    month    = month(publish_date, label = TRUE, abbr = FALSE),
    year     = year(publish_date))%>%
  select(news_provider, headline_category,day_name,year, everything(), -publish_date) #Ignore
the Publish date
```

Explanation: These are the code chunks where I concentrate on cleaning up all data before answering the questions below to create a consistent answer. It includes: - The headline_category column had untidy data such as “news.politics.oireachtas”. I used separate_longer_delim to separate those data points delimited by “.” and convert them into new rows. These new rows represent unique items in the data set. For example, one article from a news provider with the headline_category “news.politics.oireachtas” is modified so that the provider now has three entries with the same title but in different headline categories. - Next, I used mutate() to modify the headline categories since I identified that some categories had the same name but different formats. I decided to unify them into one format (e.g., “&” becomes “and”). Blank values or single spaces were converted to NA for further cleaning. - Filter() was used to ignored NA and “…” values. - Special values like “x86%” were ignored as well. - Some engagement scores were out of range, so they were ignored to concentrate strictly on the range from 0 to 1. - Lastly, I decomposed the publish_date into three smaller columns: day_name (e.g., Monday, Saturday), month, and year, and then used select() to ignore the original publish_date column.

Question 1

```
# Find the unique value of the headline_category column
clean_ireland_news %>% summarise(unique_headline_cat = n_distinct(headline_category))
```

```
## # A tibble: 1 × 1
##   unique_headline_cat
##                 <int>
## 1                 109
```

Explanation:To find the unique values within the headline category, I used: - summarise(): To perform and display the internal mathematical calculation. - n_distinct(): To identify the count of unique values within the column. To determine these values, I followed specific steps to modify and select the appropriate data as outlined in the data section.

Question 2:

Part A: Highest mean engagement

```
# Looking for provider with the highest mean engagement
clean_ireland_news %>% group_by(news_provider)%>% summarise(engagement_score = mean(engagemen
t_score)) %>% slice_max(engagement_score, n=1)
```

```
## # A tibble: 1 × 2
##   news_provider     engagement_score
##   <chr>                        <dbl>
## 1 Galway Advertiser            0.977
```

Explanation:To find the highest mean engagement scores, I used: - group_by(): To categorize the data by news_provider. - summarise(): To calculate the average engagement scores for each provider. - slice_max(): To identify and display only the provider with the highest engagement score.

Part B: Lowest articles number

```
# Looking for headline_category with the lowest articles number
clean_ireland_news %>%
count(headline_category, name = "number_of_articles") %>% slice_min(number_of_articles, n =
1)
```

```
## # A tibble: 11 × 2
##    headline_category        number_of_articles
##    <chr>                                 <int>
##  1 business-economy                          1
##  2 culture-books                             1
##  3 entertainment                             1
##  4 eye on nature                             1
##  5 lifestyle-fashion                         1
##  6 lifestyle-travel-ireland                  1
##  7 my holidays                               1
##  8 news-ireland                              1
##  9 opinion-letters                           1
## 10 report                                    1
## 11 retails-and-services                      1
```

Explanation: To find the headline_categories with the lowest number of articles, I used: - count(): To calculate the total number of articles for each headline category (counting each row, as each article is associated with one category). - slice_min(): To identify and display the categories that contain the lowest volume of articles.

Question 3:

```
# Pre-processed data table for the news provider
provider_headline_articles <- clean_ireland_news %>%
count(news_provider,headline_category, name = "number_of_articles")
```

Explanation: To find the total number of articles for each headline category and news provider, I used: - count(): To calculate the number of articles for each combination of headline category and news provider (counting each row, as each entry represents one article associated with a specific category).

```
#compute and display the statistical information, you need to use a single R function:
# Apply the summary function to total_articles, grouped by news_provider
tapply(provider_headline_articles$number_of_articles, provider_headline_articles$news_provide
r, summary)
```

```
## $`Galway Advertiser`
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##       1       1       1       1       1       1
##
## $`Irish Examiner`
##     Min.  1st Qu.   Median     Mean  3rd Qu.     Max.
##      1.0    314.5    962.5   6071.8   3197.0 198945.0
##
## $`Irish Independent`
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##       5      61     198    1226     618   40095
##
## $`Irish Times`
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##       1     419    1357    8252    4298  280056
##
## $`RTE News`
##     Min.  1st Qu.   Median     Mean  3rd Qu.     Max.
##      1.0    212.8    741.0   4640.2   2455.8 158732.0
##
## $TheJournal.ie
##     Min.  1st Qu.   Median     Mean  3rd Qu.     Max.
##      1.0    173.2    578.0   3569.5   1877.2 119494.0
```

Explanation:To find the statistical information, I used: - tapply(): To apply the summary function across the number_of_articles column while grouping the results by news_provider. This allows for a concise statistical breakdown of article counts for each individual provider in a single step.

Question 4:

Part A TheJournal.ie’s maximum engagement score?

```
# Looking for year that TheJournal.ie record its maximum engagement score
clean_ireland_news %>% filter(news_provider == "TheJournal.ie") %>% group_by(year)%>% summari
se(TheJournalie_engagement_score = sum(engagement_score)) %>% slice_max(TheJournalie_engageme
nt_score, n=1)
```

```
## # A tibble: 1 × 2
##    year TheJournalie_engagement_score
##   <dbl>                         <dbl>
## 1  2016                        12122.
```

Explanation: To find the year that TheJournal.ie record its maximum engagement score, I used: - filter(): Helps to focus on the “TheJournal.ie. - group_by(): Focus on the”year”. - summarise(): Display the answer of the sum(). - sum(): Add all the engagement scores. - slicemax() to find the only one highest engagement scores.

Part B TheJournal.ie’s maximum engagement score?

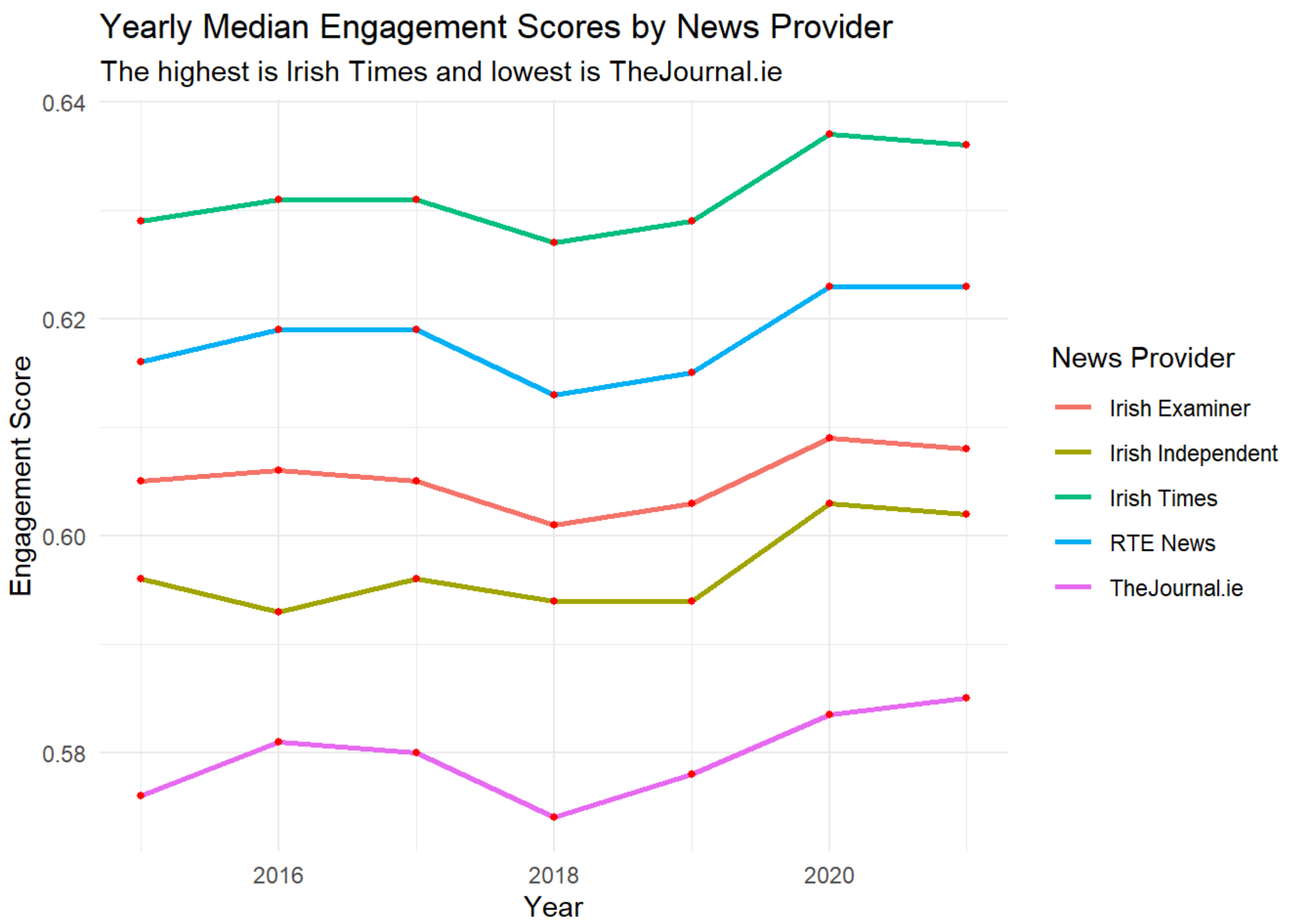
```
# News provider shows the largest increase in engagement score over the years (last - first y
ear)
provider_year_engagment_scores <- clean_ireland_news %>% group_by(news_provider,year)%>% summ
arise(total_engagement_score = sum(engagement_score), .groups = "drop") %>% arrange(year)
 # Keep the largest increase
provider_year_engagment_scores %>% group_by(news_provider)%>% summarise(first_last_score = la
st(total_engagement_score)-first(total_engagement_score)) %>% slice_max(first_last_score, n=
1)
```

```
## # A tibble: 1 × 2
##   news_provider first_last_score
##   <chr>                    <dbl>
## 1 Irish Times              4138.
```

Explanation: To find the largest increase in engagement scores over the years, I calculated the total engagement scores for each year and arranged them in chronological order starting from 1996. I then subtracted the value of the first year from the value of the last year in the dataset and used slice_max() to identify the largest increase.

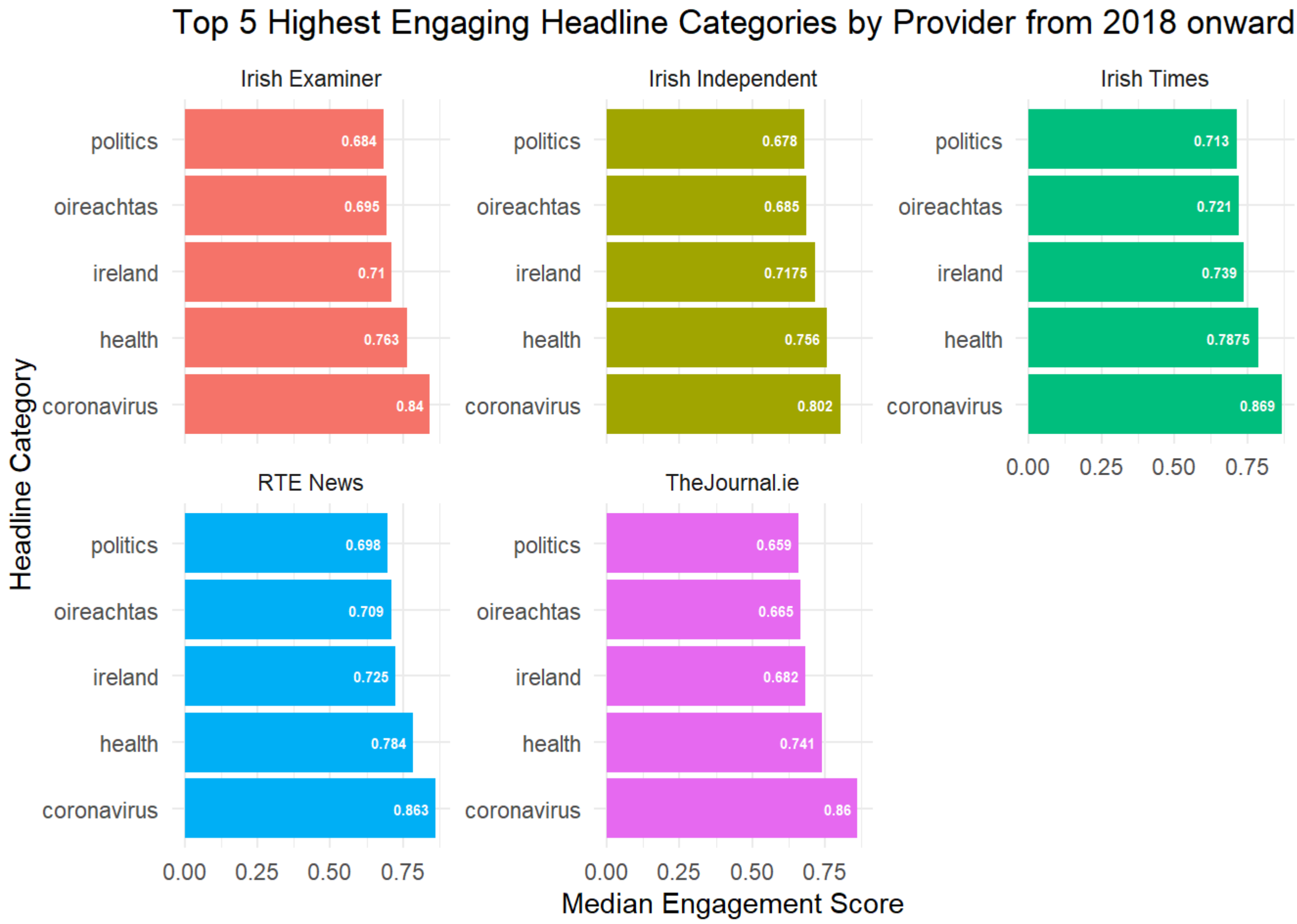
Question 5:

```
clean_ireland_news %>% group_by(news_provider, year)%>%
  # Grabs everything from 2015 onward
  filter(year >= 2015) %>% summarise(median_engagement_score = median(engagement_score)) %>%
ungroup() %>%
  ggplot(aes(x = year, y = median_engagement_score, colour= news_provider)) +

  # geom_line() creates the trend line!
  geom_line(size = 1) +
  geom_point(color = "red", size = 1) +

  # Add labels
  labs(
    title = "Yearly Median Engagement Scores by News Provider",
    subtitle = "The highest is Irish Times and lowest is TheJournal.ie",
    x = "Year",
    y = "Engagement Score",
    color = "News Provider"
  ) +
  theme_minimal()
```



Explanation: The line graph illustrates a clear ‘U-shaped’ trend in audience engagement. Between 2016 and 2018, engagement scores saw a steady decline across most providers. However, from 2018 onwards, there is a sharp and sustained increase. While the onset of the COVID-19 pandemic in 2020 provided a massive boost to engagement due to the public’s urgent need for health information during lockdown, it is notable that engagement remained high after 2020 This suggests that the pandemic may have permanently shifted Irish consumption habits toward digital news sources.

```
clean_ireland_news %>% group_by(news_provider, headline_category) %>%  filter(year >= 2018) %
>%
  # Calculate the MEDIAN engagement
  summarise(median_engagement = median(engagement_score), .groups = "drop") %>%
  # Group by provider to find the top 3 for each
  group_by(news_provider) %>%
  slice_max(order_by = median_engagement, n = 5) %>%
  ungroup() %>%

  #Plot the boxplot
  ggplot(aes(x = median_engagement, y = headline_category, fill = news_provider)) +

  geom_col(show.legend = FALSE) +

  geom_text(aes(label = median_engagement),
            hjust = 1.2,
            color = "white",
            size = 2,
            fontface = "bold") +

  facet_wrap(~ news_provider, scales = "free_y") +

  labs(
    title = "Top 5 Highest Engaging Headline Categories by Provider from 2018 onward",
    x = "Median Engagement Score",
    y = "Headline Category"
  ) +
  theme_minimal()
```



Explanation: By analyzing the top five categories per provider, we see that 'Health' and 'COVID-19' are not just popular—they are the primary drivers of engagement across almost all major Irish outlets in this period. Interestingly, while legacy providers like RTE show high engagement in 'Politics' and 'Society,' other providers might show higher scores in 'Lifestyle' or 'Entertainment.' This indicates that engagement is not just driven by the topic itself, but by the alignment between the provider's brand and the headline category. The high scores for 'Housing' and 'Cost of Living' reflect the dominant economic concerns in Ireland during this decade.

Question 6:

```
# Summarize all column into table:
news_headline_engagement <- clean_ireland_news %>% group_by(news_provider, headline_category,
year) %>%
  summarise(
    article_volume = n(),# Counts how many articles were published
    yearly_mean_engagement= mean(engagement_score),
  ) %>% ungroup()
```

Explanation: This code chunks aims to settup the question where it group into: news_provider, headline_category, year.

Part A

```r
# Question 6A
news_headline_engagement %>% group_by(news_provider, headline_category) %>%
  # Calculate the correlation
  summarise(
    association_score = cor(article_volume, yearly_mean_engagement),
    .groups = "drop"
  )  %>%
group_by(news_provider) %>%
  slice_max(abs(association_score), n = 3) %>% ungroup() #Ensure to focus on the quantity ins
tead of the sign
```

```
## # A tibble: 18 × 3
##    news_provider     headline_category     association_score
##    <chr>             <chr>                             <dbl>
##  1 Galway Advertiser business                         NA
##  2 Galway Advertiser news                             NA
##  3 Galway Advertiser sport                            NA
##  4 Irish Examiner    coronavirus                       1
##  5 Irish Examiner    bonds                            -0.897
##  6 Irish Examiner    construction                      0.858
##  7 Irish Independent bonds                            -1
##  8 Irish Independent generation-emigration             0.980
##  9 Irish Independent equities                          0.950
## 10 Irish Times       coronavirus                       1
## 11 Irish Times       generation-emigration             0.978
## 12 Irish Times       working-abroad                   -0.954
## 13 RTE News          coronavirus                       1
## 14 RTE News          tuarascail                       -0.977
## 15 RTE News          work                              0.913
## 16 TheJournal.ie     coronavirus                       1
## 17 TheJournal.ie     africa                           -0.894
## 18 TheJournal.ie     food                              0.879
```

Explanation: To identify the top three headline categories with the strongest association between yearly article volume and yearly mean engagement scores, I used the cor() function. I applied the abs() function to focus on the magnitude or strength of the association, as the largest absolute values indicate the strongest relationships regardless of direction. I chose to retain the Galway Advertiser as NA when reporting association scores, despite the lack of available values. This demonstrates that an association may not exist for this provider, confirming that the Galway Advertiser did not have any headline categories with a strong association.

Part B:

```r
# Question 6B
news_headline_engagement %>% group_by(news_provider, headline_category) %>%

  # Calculate the correlation
  summarise(
    increase_trends = cor(year,article_volume),
    .groups = "drop"
  )  %>%
group_by(news_provider) %>%
  slice_max(increase_trends, n = 3) %>% ungroup() #Ensure to focus on the positive sign
```

```
## # A tibble: 18 × 3
##    news_provider     headline_category increase_trends
##    <chr>             <chr>                       <dbl>
##  1 Galway Advertiser business                   NA
##  2 Galway Advertiser news                       NA
##  3 Galway Advertiser sport                      NA
##  4 Irish Examiner    restaurant                  0.899
##  5 Irish Examiner    parenting                   0.853
##  6 Irish Examiner    ireland                     0.826
##  7 Irish Independent bonds                       1
##  8 Irish Independent lifestyle                   0.832
##  9 Irish Independent health-family               0.814
## 10 Irish Times       tuarascail                  0.907
## 11 Irish Times       ireland                     0.812
## 12 Irish Times       lifestyle                   0.811
## 13 RTE News          lifestyle                   0.821
## 14 RTE News          health-family               0.781
## 15 RTE News          ireland                     0.772
## 16 TheJournal.ie     lifestyle                   0.811
## 17 TheJournal.ie     ireland                     0.801
## 18 TheJournal.ie     europe                      0.764
```

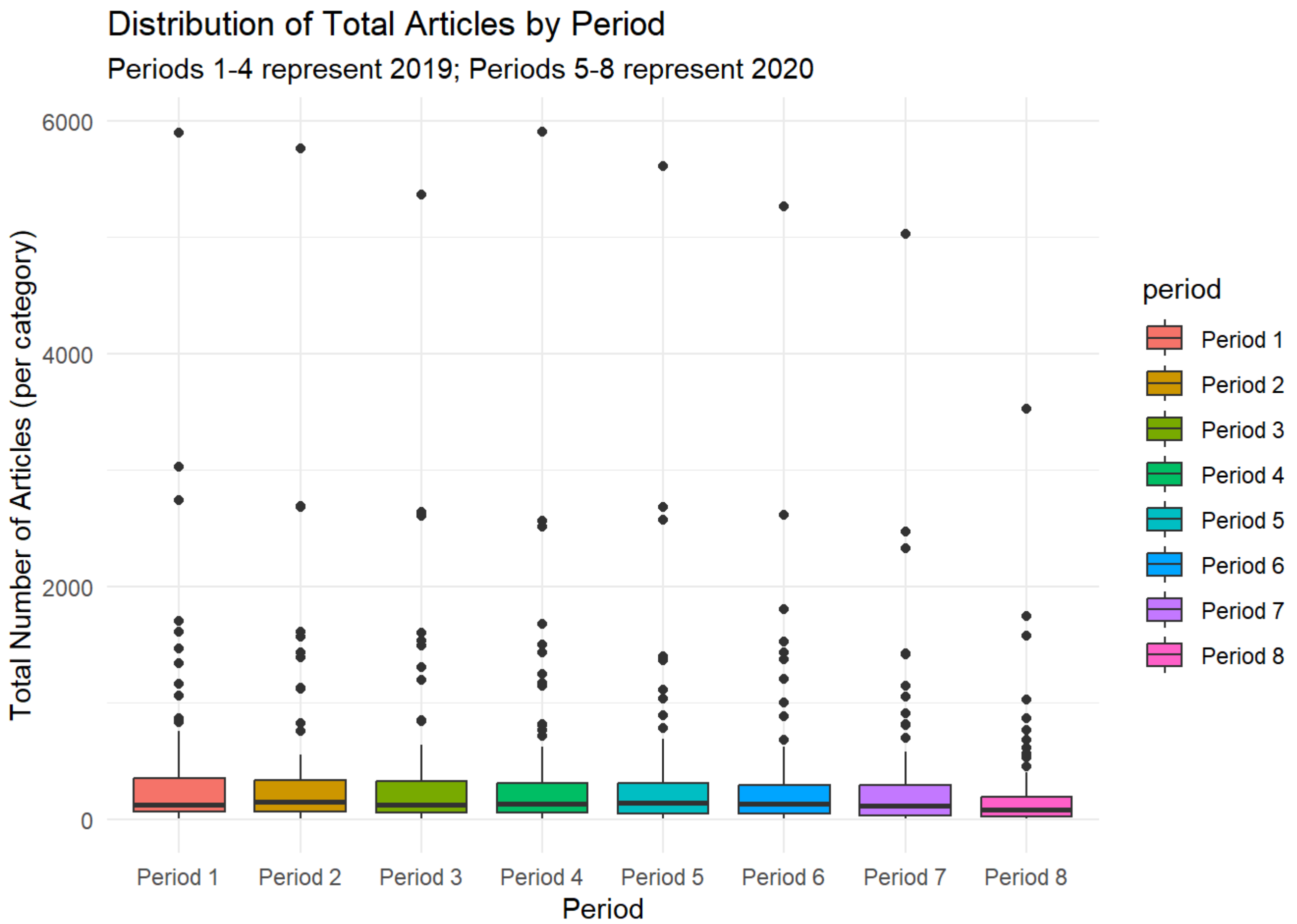
Explanation: Correlation demonstrates the relationship between two variables based on their strength and direction. Building on the previous question, I chose to focus exclusively on the largest positive correlation values to identify the top three headline categories with the most significant increasing trends in yearly article counts.

Question 7:

```r
# Determine the period
period_data <- clean_ireland_news %>%
  # 1. Filter to keep only the years you need
  filter(year == 2019 | year == 2020) %>%

  # 2. Create the Period column using just the Year and Month
  mutate(
    period = case_when(
      # 2019 Periods
      year == 2019 & month %in% c("January", "February", "March")    ~ "Period 1",
      year == 2019 & month %in% c("April", "May", "June")   ~ "Period 2",
      year == 2019 & month %in% c("July", "August", "September")    ~ "Period 3",
      year == 2019 & month %in% c("October", "November", "December") ~ "Period 4",

      # 2020 Periods
      year == 2020 & month %in% c("January", "February", "March")    ~ "Period 5",
      year == 2020 & month %in% c("April", "May", "June")   ~ "Period 6",
      year == 2020 & month %in% c("July", "August", "September")   ~ "Period 7",
      year == 2020 & month %in% c("October", "November", "December") ~ "Period 8"
    )
)
```

Explanation: To assign the period, I realize that each period related to the quarter of that year so I “hard-code” to assign period’s value to the columns.

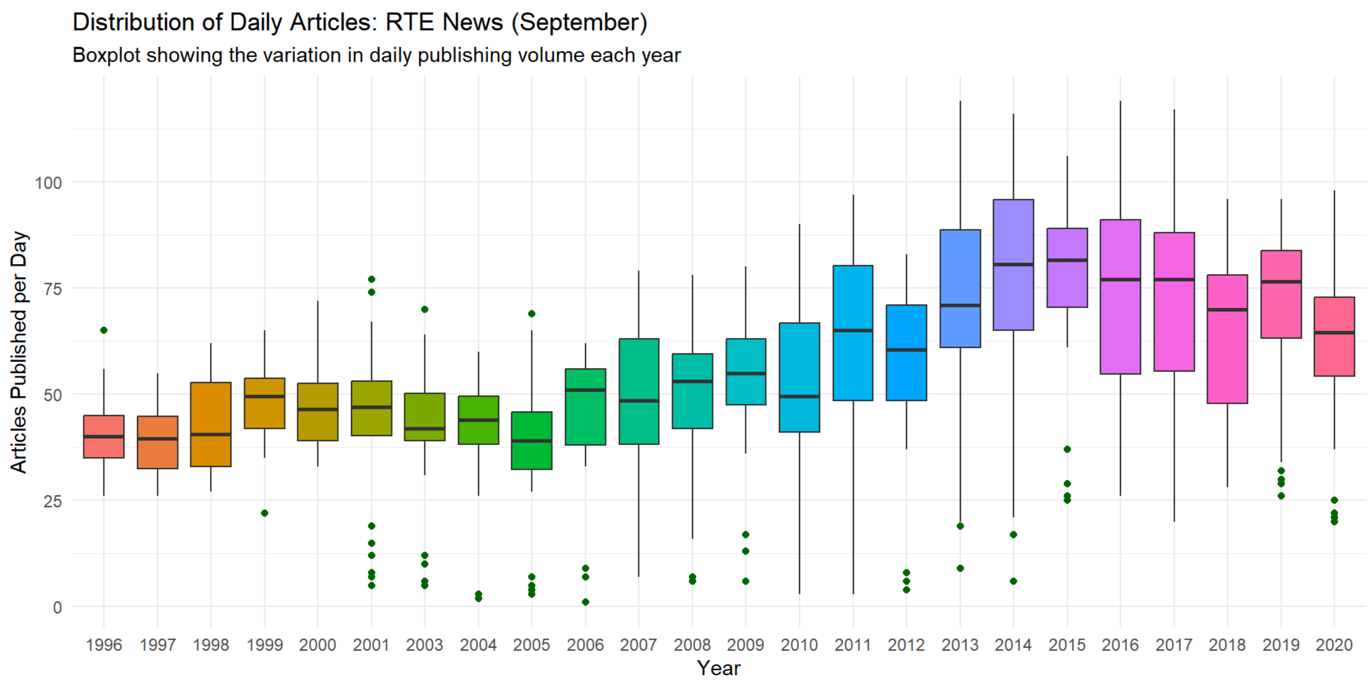
```
# Calculate the total period
period_data %>% group_by(period, headline_category) %>%
  summarise(
    period_total_articles = n() # Computes the total count of articles
  ) %>% ungroup() %>%
  ggplot(aes(x = period, y = period_total_articles, fill = period)) +
  # Generate the boxplot
  geom_boxplot(alpha = 1.0, show.legend = TRUE) +

  # Add labels to make it professional
  labs(
    title = "Distribution of Total Articles by Period",
    subtitle = "Periods 1-4 represent 2019; Periods 5-8 represent 2020",
    x = "Period",
    y = "Total Number of Articles (per category)"
  ) +
  theme_minimal()
```



Explanation: From the finding above, I use geom_boxplot() to draw the geom_boxplot() where x is period and y is the total amount of Articles.

Question 8:

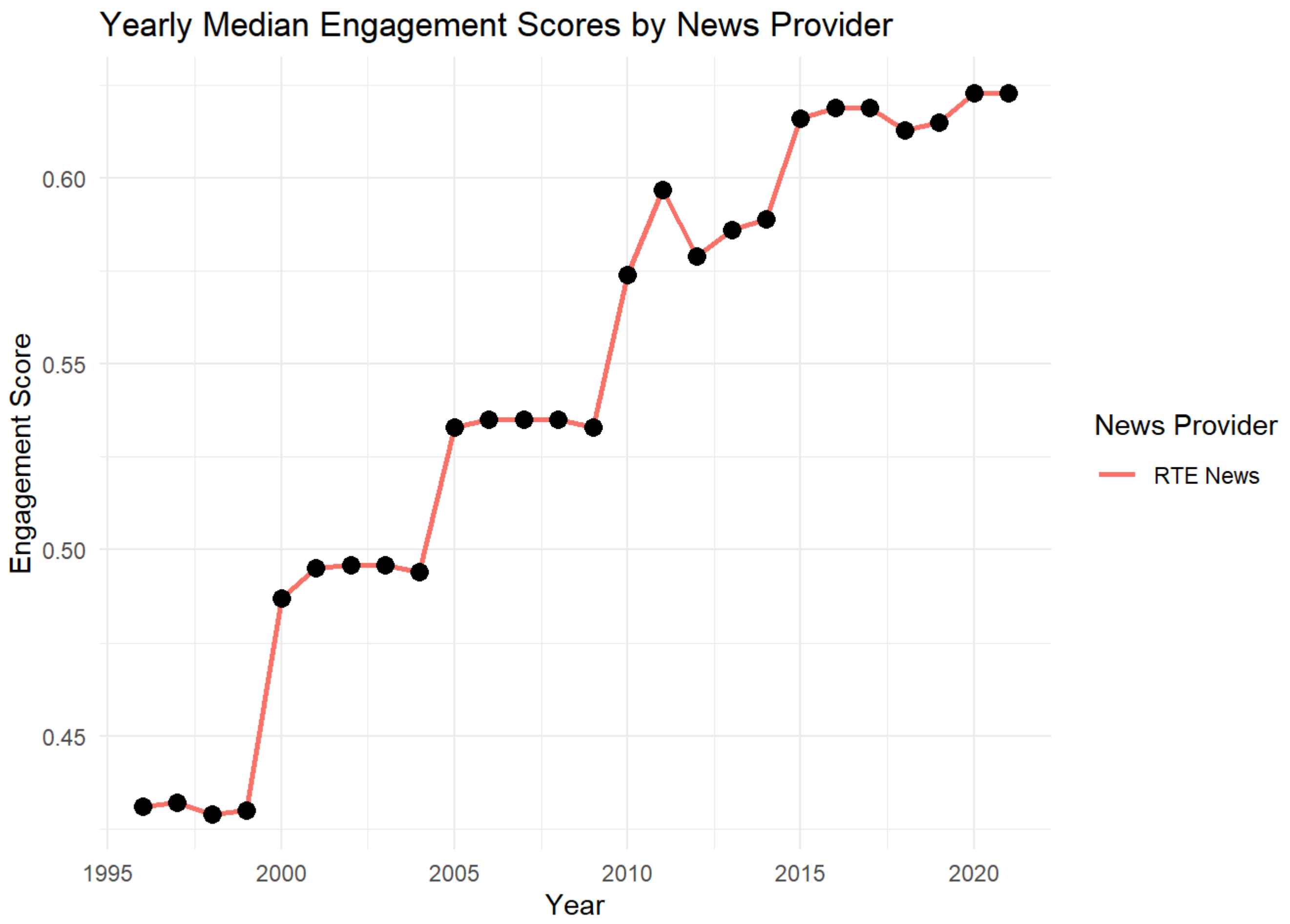
```
clean_ireland_news %>%
  # 1. Focus ONLY on RTE in September
  filter(news_provider == "RTE News" & month == "September") %>%
  # 2. Count articles per day so the boxplot has 'spread' to show
  group_by(year, date) %>%
  summarise(daily_articles = n()) %>% ungroup() %>%

  # 3. Plotting
  ggplot(aes(x = factor(year), y = daily_articles, fill = factor(year))) +

  geom_boxplot(show.legend = FALSE, outlier.color = "darkgreen") +

  labs(
    title = "Distribution of Daily Articles: RTE News (September)",
    subtitle = "Boxplot showing the variation in daily publishing volume each year",
    x = "Year",
    y = "Articles Published per Day"
  ) +
  theme_minimal()
```



Explanation: I chose to use a boxplot because, unlike other graph types, it effectively demonstrates the distribution of daily article counts for each September across multiple years. This visualization provides a clear comparison between years, making it easier to identify variations in publishing volume.

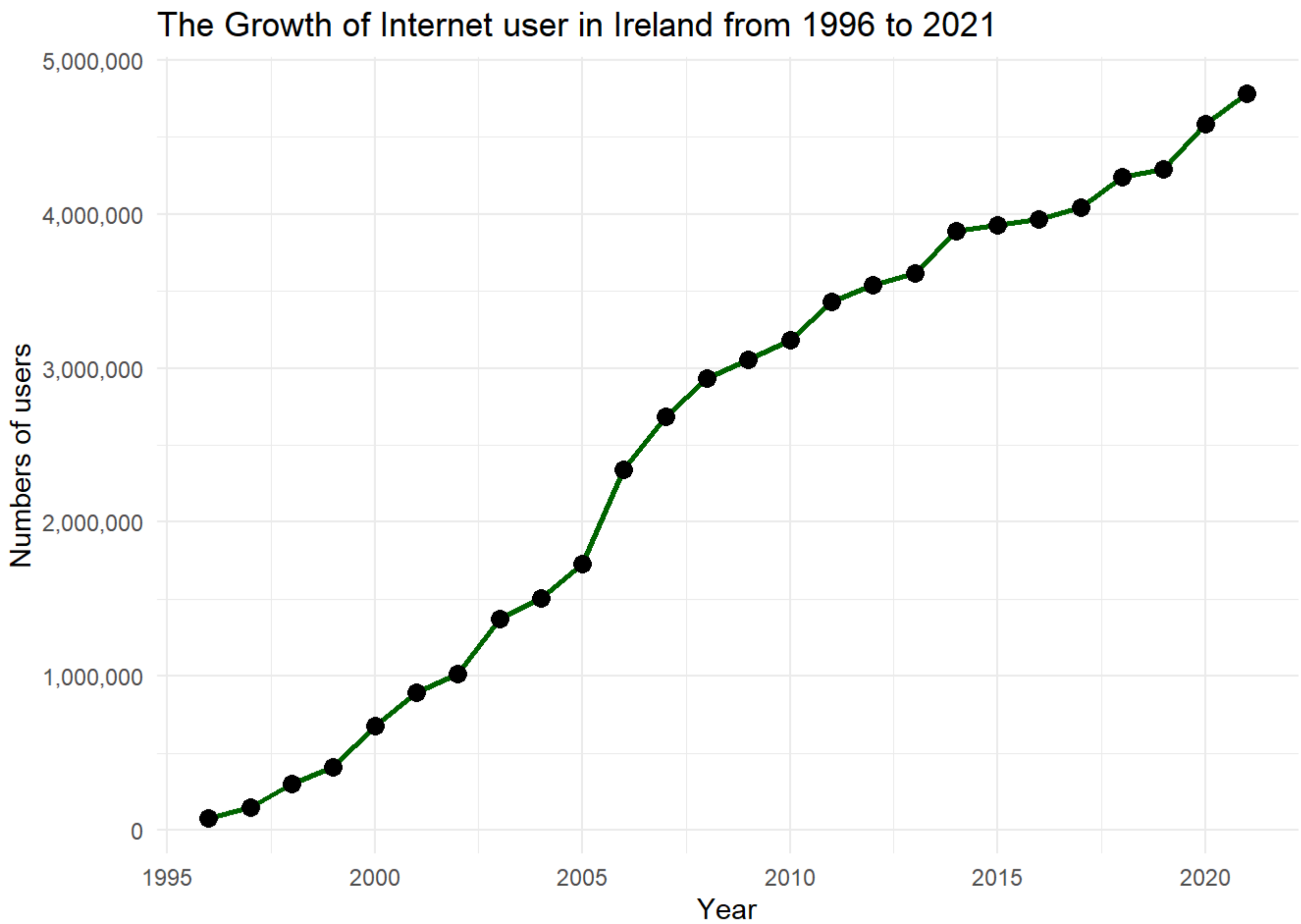
Question 9:

```
clean_ireland_news %>% group_by(news_provider, year) %>%  filter(news_provider == "RTE News")
%>% summarise(median_engagement_score = median(engagement_score)) %>% ungroup() %>%
  ggplot(aes(x = year, y = median_engagement_score, colour= news_provider)) +

  # geom_line() creates the trend line!
  geom_line(size = 1) +
  geom_point(color = "black", size = 3) +

  # Add labels
  labs(
    title = "Yearly Median Engagement Scores by News Provider",
    x = "Year",
    y = "Engagement Score",
    color = "News Provider"
  ) +
  theme_minimal()
```



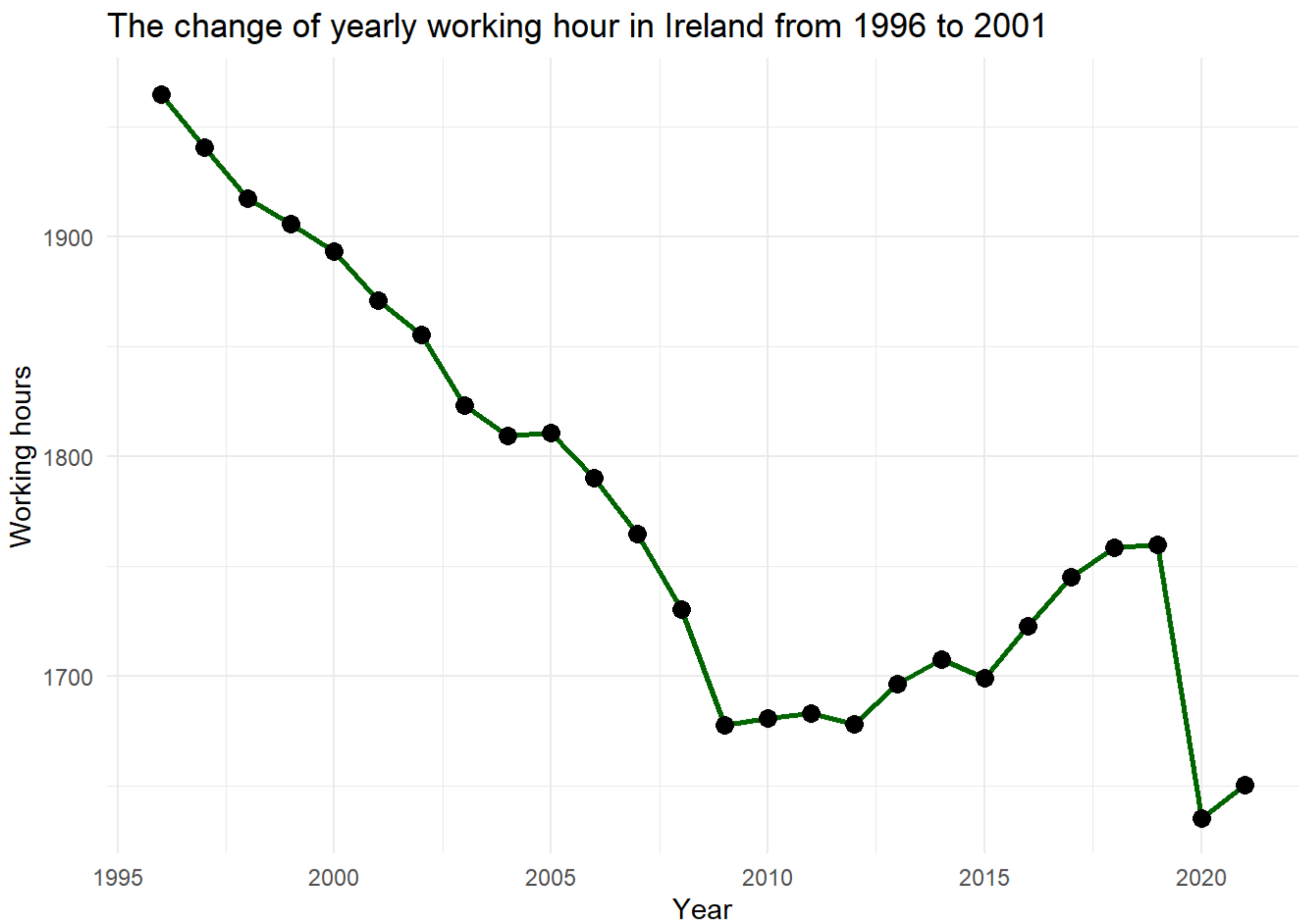
Explanation: Overall, this graph demonstrates the increase in engagement for RTE News. I focused on the period between 1996 and 2005, as this timeframe coincides with the emergence and development of the internet, as well as a simultaneous reduction in working hours.

```
internet_users %>% group_by(Entity, Year) %>% filter(Entity == "Ireland" & between(Year, 199
6, 2021)) %>% summarise(Number.of.Internet.users = sum(Number.of.Internet.users)) %>% ungroup
() %>%
  ggplot(aes(x = Year, y = Number.of.Internet.users)) +

  # geom_line() creates the trend line!
  geom_line(size = 1, color="darkgreen") +
  geom_point(color = "black", size = 3) +

  #Scale the y to make them look neatly
  scale_y_continuous(labels = scales::comma) +

  # Add labels
  labs(
    title = "The Growth of Internet user in Ireland from 1996 to 2021",
    x = "Year",
    y = "Numbers of users"
  ) +
  theme_minimal()
```



Explanation: From 1996 to 2005, there was a notable surge in the number of internet users, increasing by more than 1 million. This growing online presence became a key factor in driving engagement scores for RTE News, as the internet allowed users to access articles from anywhere and at any time.

```
working_hour %>% group_by(Entity, Year) %>% filter(Entity == "Ireland" & between(Year, 1996,
2021)) %>%
  ggplot(aes(x = Year, y = `Working hours per worker`)) +

  # geom_line() creates the trend line!
  geom_line(size = 1, color="darkgreen") +
  geom_point(color = "black", size = 3)+

  # Add labels
  labs(
    title = "The change of yearly working hour in Ireland from 1996 to 2001 ",
    x = "Year",
    y = "Working hours"
  ) +
  theme_minimal()
```



Explanation: From 1996 to 2005, yearly working hours in Ireland decreased sharply, with a particularly significant decline occurring between 1996 and 2010. This trend suggest that people had more leisure time for other activities, including reading news articles. Although there was a slight increase in working hours from 2010 to 2019, the reading habits established during the previous period likely persisted, ensuring that engagement scores did not drop.